

Normal Distribution Analysis using Python

A company claims that the IQ scores of its employees are normally distributed with a mean of 100 and a standard deviation of 15.

Write a Python program to:

1. Generate 1000 random IQ scores based on the given distribution.
2. Plot a histogram of the IQ scores and overlay the Probability Density Function (PDF).
3. Calculate and display the mean, median, and standard deviation of the generated data. Compare them with the theoretical values (100, 15).
4. Using `scipy.stats.norm`, compute the probability that an employee's IQ is:
 - Less than 90
 - Greater than 110
 - Between 95 and 105
5. Plot the Cumulative Distribution Function (CDF) for IQ values ranging from 60 to 140.